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MariaDB [indassignment2]> CREATE DATABASE MainAssignment;

Query OK, 1 row affected (0.001 sec)

MariaDB [indassignment2]> USE MainAssignment;

Database changed

MariaDB [MainAssignment]> -- 1. Base Table: Department

MariaDB [MainAssignment]> CREATE TABLE Department (

-> deptNo VARCHAR(5) PRIMARY KEY,

-> deptName VARCHAR(50) NOT NULL,

-> mgrEmpNo VARCHAR(5)

->);

Query OK, 0 rows affected (0.010 sec)

MariaDB [MainAssignment]> -- 2. Base Table: Employee

MariaDB [MainAssignment]> CREATE TABLE Employee (

-> empNo VARCHAR(5) PRIMARY KEY,

-> fName VARCHAR(30) NOT NULL,

-> lName VARCHAR(30) NOT NULL,

-> address VARCHAR(100),

-> DOB DATE,

-> sex CHAR(1) NOT NULL CHECK (sex IN ('M', 'F')),

-> position VARCHAR(30) NOT NULL CHECK (position IN ('Manager', 'Team Leader',
'Analyst', 'Software Developer')),

-> deptNo VARCHAR(5),

-> FOREIGN KEY (deptNo) REFERENCES Department(deptNo)

->);

Query OK, 0 rows affected (0.007 sec)

MariaDB [MainAssignment]> ALTER TABLE Department ADD FOREIGN KEY (mgrEmpNo)
REFERENCES Employee(empNo);

Query OK, 0 rows affected (0.015 sec)

Records: 0 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]>

MariaDB [MainAssignment]> -- 3. Base Table: Project

MariaDB [MainAssignment]> CREATE TABLE Project (

-> projNo VARCHAR(5) PRIMARY KEY,
-> projName VARCHAR(50) NOT NULL,
-> deptNo VARCHAR(5) NOT NULL,
-> FOREIGN KEY (deptNo) REFERENCES Department(deptNo)
->);

Query OK, 0 rows affected (0.007 sec)

MariaDB [MainAssignment]> -- 4. Base Table: WorksOn

MariaDB [MainAssignment]> CREATE TABLE WorksOn (

-> empNo VARCHAR(5),
-> projNo VARCHAR(5),
-> dateWorked DATE,
-> hoursWorked INT CHECK (hoursWorked >= 0 AND hoursWorked <= 40),
-> PRIMARY KEY (empNo, projNo, dateWorked),
-> FOREIGN KEY (empNo) REFERENCES Employee(empNo),
-> FOREIGN KEY (projNo) REFERENCES Project(projNo)
->);

Query OK, 0 rows affected (0.006 sec)

MariaDB [MainAssignment]> CREATE TABLE Part (

```
-> partNo VARCHAR(10),  
-> contract VARCHAR(20),  
-> partCost DECIMAL(10, 2) NOT NULL,  
-> PRIMARY KEY (partNo, contract)  
-> );
```

Query OK, 0 rows affected (0.005 sec)

MariaDB [MainAssignment]> -- Insert Departments

MariaDB [MainAssignment]> INSERT INTO Department (deptNo, deptName, mgrEmpNo)
VALUES

```
-> ('D1', 'Information Technology', NULL),  
-> ('D2', 'Planning', NULL),  
-> ('D3', 'Research & Development', NULL);
```

Query OK, 3 rows affected (0.001 sec)

Records: 3 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]> INSERT INTO Employee (empNo, fName, lName, address, DOB,
sex, position, deptNo) VALUES

```
-> ('E1', 'Sarah', 'Connor', '123 Tech Lane', '1992-05-14', 'F', 'Manager', 'D1'),  
-> ('E2', 'John', 'Doe', '456 Cyber Ave', '1994-10-22', 'M', 'Software Developer', 'D1'),  
-> ('E3', 'Ellen', 'Ripley', '789 Orbit Way', '1988-01-05', 'F', 'Manager', 'D2'),  
-> ('E4', 'James', 'Smith', '101 Plan Blvd', '1995-07-19', 'M', 'Analyst', 'D2'),  
-> ('E5', 'Marcus', 'Wright', '202 Forge St', '1991-11-30', 'M', 'Team Leader', 'D3');
```

Query OK, 5 rows affected (0.001 sec)

Records: 5 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]> UPDATE Department SET mgrEmpNo = 'E1' WHERE deptNo = 'D1';

Query OK, 1 row affected (0.001 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [MainAssignment]> UPDATE Department SET mgrEmpNo = 'E3' WHERE deptNo = 'D2';

Query OK, 1 row affected (0.001 sec)

Rows matched: 1 Changed: 1 Warnings: 0

MariaDB [MainAssignment]>

MariaDB [MainAssignment]> -- Insert Projects (Explicitly creating 'SCCS')

MariaDB [MainAssignment]> INSERT INTO Project (projNo, projName, deptNo) VALUES

-> ('SCCS', 'Security Core Control System', 'D1'),

-> ('P102', 'Cloud Migration Sync', 'D1'),

-> ('P201', 'Strategic Horizon 2030', 'D2'),

-> ('P301', 'Alpha Engine Prototype', 'D3');

Query OK, 4 rows affected (0.001 sec)

Records: 4 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]> -- Insert WorksOn allocation entries

MariaDB [MainAssignment]> INSERT INTO WorksOn (empNo, projNo, dateWorked, hoursWorked) VALUES

-> ('E1', 'SCCS', '2026-06-01', 8),

-> ('E1', 'P102', '2026-06-02', 6),

-> ('E2', 'SCCS', '2026-06-01', 8),

-> ('E2', 'SCCS', '2026-06-02', 7),

-> ('E3', 'P201', '2026-06-01', 8),

-> ('E4', 'P201', '2026-06-01', 5),

-> ('E5', 'P301', '2026-06-01', 10);

Query OK, 7 rows affected (0.001 sec)

Records: 7 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]> INSERT INTO Part (partNo, contract, partCost) VALUES

-> ('PT-01', 'CONT-A', 1250.00),

-> ('PT-01', 'CONT-B', 950.00),

-> ('PT-02', 'CONT-A', 450.00),

-> ('PT-03', 'CONT-C', 2100.00);

Query OK, 4 rows affected (0.001 sec)

Records: 4 Duplicates: 0 Warnings: 0

MariaDB [MainAssignment]>

MariaDB [MainAssignment]> CREATE VIEW FemaleManagedProjects AS

-> SELECT p.projNo, p.projName, p.deptNo

-> FROM Project p

-> JOIN Department d ON p.deptNo = d.deptNo

-> JOIN Employee e ON d.mgrEmpNo = e.empNo

-> WHERE e.sex = 'F'

-> ORDER BY p.projNo;

Query OK, 0 rows affected (0.004 sec)

MariaDB [MainAssignment]>

MariaDB [MainAssignment]> CREATE VIEW EmployeeProjectHours AS

-> SELECT e.empNo, e.fName, e.lName, p.projName, w.hoursWorked

-> FROM Employee e

-> JOIN WorksOn w ON e.empNo = w.empNo

-> JOIN Project p ON w.projNo = p.projNo;

Query OK, 0 rows affected (0.003 sec)

MariaDB [MainAssignment]>

MariaDB [MainAssignment]> CREATE VIEW EmpProject(empNo, projNo, totalHours)

-> AS SELECT w.empNo, w.projNo, SUM(hoursWorked)

-> FROM Employee e, Project p, WorksOn w

-> WHERE e.empNo = w.empNo AND p.projNo = w.projNo

-> GROUP BY w.empNo, w.projNo;

Query OK, 0 rows affected (0.003 sec)

MariaDB [MainAssignment]> SELECT * FROM EmpProject;

empNo	projNo	totalHours
E1	P102	6
E1	SCCS	8
E2	SCCS	15
E3	P201	8
E4	P201	5
E5	P301	10

6 rows in set (0.001 sec)

MariaDB [MainAssignment]> SELECT projNo FROM EmpProject WHERE projNo = 'SCCS';

projNo
SCCS
SCCS

+-----+

2 rows in set (0.000 sec)

MariaDB [MainAssignment]> SELECT COUNT(projNo) FROM EmpProject WHERE empNo = 'E1';

+-----+

| COUNT(projNo) |

+-----+

| 2 |

+-----+

1 row in set (0.000 sec)

MariaDB [MainAssignment]> SELECT empNo, totalHours FROM EmpProject GROUP BY empNo;

+-----+-----+

| empNo | totalHours |

+-----+-----+

| E1 | 6 |

| E2 | 15 |

| E3 | 8 |

| E4 | 5 |

| E5 | 10 |

+-----+-----+

5 rows in set (0.001 sec)

MariaDB [MainAssignment]>